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Project Report On

"WazirX Cyber Attack : A Cybersecurity Case Study(2024)"

Submitted in partial fulfilment of the requirement for the project work for the degree of

BACHELOUR OF COMPUTER APPLICATION (BCA)

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BANGALORE UNIVERSITY

Under the Guidance Of

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CERTIFICATE

This is to certify that the project synopsis entitled “WazirX Cyber Attack: A Cybersecurity Case Study(2024)” is submitted by Ms Srilekha v(USN: U03BX24S0095) in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) from Bangalore University for the academic year 2025–2026.

The project synopsis has been prepared under the guidance and supervision of the undersigned. The synopsis is submitted for academic evaluation in accordance with the applicable department and university guidelines.

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ABSTRACT

This case study is about the cyber attack on WazirX, an Indian cryptocurrency exchange, reported on 18 July 2024. The incident became a major cybersecurity concern because a large amount of cryptocurrency was lost from one of its digital wallets.

According to public reports, the loss was more than US$230 million. WazirX stated that the incident involved a multi-signature wallet and raised concerns about the transaction approval process. WazirX and its custody provider, Liminal, gave different explanations about the security failure.

This report explains the background of WazirX,the incident,wallet security,the reported attack mechanism and its financial impact.It also discusses the response to the incident and the lessons that other organizations can learn from it.

The main purpose of this study is to understand why strong security controls, proper transaction verification and regular security checks are important.The report is based on publicly available information and presents the incident for academic study.

INTRODUCTION

Cybersecurity has become an important part of our daily lives because people and organisations depend on digital technology for communication, banking, shopping and business activities. As the use of online services increases, the risk of cyber attacks also increases.

A cyber attack is an attempt to access, damage, disrupt or steal information or digital assets without proper permission. Such attacks can affect individuals, companies and even government organisations. Financial services are particularly important to protect because a security failure can lead to monetary loss and a loss of customer trust.

Cryptocurrency exchanges allow users to buy, sell and store digital currencies. These platforms need strong security systems to protect users' assets. Unlike some traditional financial transactions, blockchain transactions may be difficult to reverse once they are confirmed.

WazirX is an Indian cryptocurrency exchange that reported a major cyber attack on 18 July 2024. The incident involved one of its multi-signature wallets, and the reported loss exceeded US$230 million. The incident also raised questions about how transactions were checked and approved.

The public explanations given by WazirX and its custody provider differed regarding the source of the security failure. Therefore, it is important to distinguish between reported facts and claims that were disputed.

This case study examines the incident from a cybersecurity point of view. It explains the importance of wallet security, transaction verification and third-party risk management.

OBJECTIVES OF THE STUDY

The main objectives of this case study are:

To understand the WazirX cyber attack reported in July 2024.

To study the importance of cryptocurrency wallet security.

To understand the purpose of multi-signature wallets.

To explain the reported attack mechanism in simple terms.

To examine the financial and operational impact of the incident.

To understand the response and recovery efforts.

To identify the security lessons from the incident.

To suggest measures that may help prevent similar incidents in the future.

RESEARCH AND METHODOLOGY

This case study is based on secondary research. Information was collected from publicly available news reports, official statements and online articles related to the WazirX cyber attack.

The collected information was organised under different headings, including the incident overview, wallet security, attack mechanism, financial impact and security recommendations.

The report uses public information to explain the incident in simple language. Since this study does not include access to private investigation records or technical evidence, it does not claim to establish the final cause of the attack independently.

BACKGROUND OF WAZIRX

WazirX is an Indian cryptocurrency exchange that allows users to trade and manage digital assets.

The exchange used a multi-signature wallet associated with Liminal, a digital asset custody provider. A multi-signature wallet requires approval from more than one authorised person before a transaction is completed.

However, multiple approvals do not guarantee complete security. The transaction details must also be checked carefully, and the approval systems must be protected.

INCIDENT OVERVIEW

On 18 July 2024, WazirX reported a cyber attack involving one of its multi-signature wallets. More than US$230 million in cryptocurrency was reportedly lost.

WazirX halted withdrawals while investigating the incident. WazirX stated that there was a mismatch between the information shown on the custody interface and the actual transaction. Liminal disputed claims that its infrastructure had been compromised.

Because the explanations differed, the exact technical cause remains disputed.

IMPORTANT EVENTS

18 July 2024: WazirX reported the attack and halted withdrawals.

Following days: WazirX and Liminal gave different explanations.

Following months: Investigation, asset tracing and legal steps continued.

August 2024: WazirX announced a moratorium application in the Singapore High Court.

WALLET SECURITY

Cryptocurrency wallet security protects digital assets from unauthorized access and transfers.

What is a Multi-Signature Wallet

A multi- signature wallet requires approval from multiple authorized people.This reduces the risk of one person controlling the funds alone.

However,signers must understand and verify every transaction before approving it.

Importance of Transaction Verification

Before approving a transaction,users should check the destination address,amount,asset and purpose. Smart-contrast transactions and permission changes should also be reviewed carefully.

Role of Third-Party Providers

External providers may manage important security

services.Organisations must regularly review their controls and ensure that transaction information is accurate throughout the approval process.

REPORTED ATTACK MECHANISM

The exact cause of the attack was disputed. According to WazirX's preliminary explanation, the transaction information shown to the signers may not have matched the actual transaction.

In simple terms:

1. A transaction was prepared.
2. Signers reviewed the displayed information.
3. The required approvals were given.
4. The transaction allowed the attacker to move cryptocurrency.

This is only a simplified explanation of the reported concern.

FINANCIAL IMPACT

WazirX reported a loss of more than US$230 million in cryptocurrency.

Customers faced uncertainty about their balances and could not withdraw funds during the restrictions. The company also faced investigation costs, legal expenses, business disruption and reputational damage.

The incident raised wider concerns about the security of cryptocurrency exchanges and digital asset custody.

RESPONSE AND RECOVERY

WazirX investigated the incident, traced the stolen cryptocurrency and communicated with customers. Blockchain records can help track funds, but tracing does not guarantee recovery.

The response also involved legal and financial steps. In August 2024, WazirX announced a moratorium application in the Singapore High Court as part of a restructuring process.

SECURITY LESSONS

The WazirX cyber attack of July 2024 highlights the importance of cybersecurity in cryptocurrency exchanges. The incident showed that even platforms that use advanced security technologies can face serious risks if their security systems and procedures are not strong enough.

One important lesson from this incident is the need for strong digital wallet security. Cryptocurrency exchanges should protect their wallets using multiple security measures and regularly review their wallet management procedures. Since cryptocurrency transactions can be difficult to reverse, preventing unauthorized transactions is extremely important.

Another lesson is the importance of secure transaction approval. Cryptocurrency platforms should use strong authentication methods, independent verification, and clear approval procedures for large or unusual transactions. These measures can help reduce the risk of unauthorized transfers.

The incident also highlights the need for regular security audits and vulnerability assessments. Security teams should examine their systems frequently to identify weaknesses before attackers can exploit them. Companies should also test their security controls and update them whenever new risks are identified.

Incident response is another important lesson. Organizations should have a clear plan for responding to cyber attacks. This plan should include detecting suspicious activity, temporarily restricting affected services when necessary, informing users, preserving evidence, and coordinating with cybersecurity experts and law enforcement authorities.

Finally, transparent communication is essential during a cybersecurity incident. Customers need accurate and timely information about what happened, which services are affected, and what steps the company is taking. Good communication can help reduce confusion and maintain trust.

RECOMMENDATIONS

The WazirX cyber attack highlights several areas where cryptocurrency exchanges can improve their security. The following recommendations may help exchanges protect digital assets and reduce the risk of similar incidents in the future.

1. Strengthen Multi-Signature Wallet Security

Cryptocurrency exchanges should use secure multi-signature wallets, where multiple authorized approvals are required before a transaction can be completed. However, multiple signatures alone may not be sufficient. Each approval should include an independent verification of the transaction details.

2. Verify Transaction Details Independently

The transaction information displayed on a website or wallet interface should match the actual transaction being signed. Signers should independently verify the destination address, amount, token, and smart contract operation before approving a transaction. Any mismatch should stop the transaction until it is investigated.

3. Improve Access Control and Authentication

Exchanges should use strong authentication methods, including multi-factor authentication, secure hardware devices, and role-based access control. Access permissions should be limited to employees who need them for their work. Unusual login attempts and changes to important security settings should trigger alerts.

4. Conduct Regular Security Audits

Independent cybersecurity experts should regularly examine wallet infrastructure, smart contracts, transaction approval systems, and third-party services. Penetration testing and vulnerability assessments can help identify security weaknesses before they are exploited.

5. Monitor Transactions in Real Time

Cryptocurrency exchanges should use automated monitoring systems to identify unusual transfers, unexpected contract changes, and suspicious transaction patterns. High-value or abnormal transactions should receive additional verification. Where possible, emergency controls should allow the organization to pause affected operations safely.

6. Strengthen Third-Party Security

Exchanges often depend on external companies for wallet infrastructure and custody services. They should assess the security practices of these providers before using their services and continue reviewing them regularly. Contracts should clearly define security responsibilities, incident reporting procedures, and cooperation during investigations.

7. Prepare an Incident Response Plan

Every exchange should maintain a written incident response plan. The plan should

explain how to identify an attack, protect remaining assets, preserve digital evidence, notify the relevant authorities, and communicate with affected users. Regular exercises can help employees respond quickly and effectively.

8. Improve Transparency and User Communication

During a cyber attack, the company should provide accurate updates through verified communication channels. It should explain confirmed facts, identify information that is still under investigation, and avoid making unsupported claims about recovery. This can help reduce confusion and protect users from false information.

9. Provide Cybersecurity Training

Employees who handle digital assets should receive regular training on phishing, social engineering, suspicious transaction requests, and secure approval practices. Training should include practical examples so that staff can recognize unusual activity and report it promptly.

10. Maintain Regular Security Reviews

Security procedures should be reviewed whenever new technologies, wallet systems, or external service providers are introduced. Exchanges should also learn from previous incidents and update their policies to address newly identified risks.

By implementing these recommendations, cryptocurrency exchanges can strengthen their security practices, improve their response to cyber threats, and reduce the possibility of major financial losses. No single security measure can eliminate every risk, so a combination of technical controls, trained staff, independent reviews, and clear procedures is essential.

CONCLUSION

The WazirX cyber attack of 18 July 2024 was a significant cybersecurity incident that affected a multi-signature cryptocurrency wallet and resulted in the loss of digital assets worth more than $230 million, according to WazirX's preliminary report. The incident raised serious concerns about the security of cryptocurrency exchanges and the protection of users' digital assets.

According to WazirX's initial findings, the attack involved a mismatch between the transaction information displayed on the wallet service interface and the actual transaction that was signed. The company reported that the incident affected one of its multi-signature wallets managed using Liminal's digital asset custody and wallet infrastructure. The exact technical details should be understood in the context of the investigation and the findings published by the organizations involved.

The incident demonstrates that using multi-signature wallets and hardware wallets

does not automatically guarantee complete security. Transaction verification, secure interfaces, independent security checks, access control, and continuous monitoring are also important. Organizations must ensure that their security systems work together effectively rather than depending on a single protective measure.

The attack also had financial and operational consequences. It affected the company's ability to maintain its normal operations and created uncertainty for users regarding their digital assets. The response included reporting the incident to relevant authorities, working with cybersecurity specialists, and attempting to trace the stolen assets.

This case study helped me understand the importance of cybersecurity in protecting digital financial systems. It also showed how a security incident can affect not only an organization but also its customers and reputation. The lessons from this incident can help cryptocurrency exchanges improve their security policies and prepare for future cyber threats.

In conclusion, strong cybersecurity requires continuous improvement, regular security audits, secure transaction approval procedures, employee awareness, and transparent communication. The WazirX incident serves as an important example of why digital asset platforms must identify potential weaknesses and take preventive measures to protect their systems and users.

REFERENCES

1. WazirX. (18 July 2024). Preliminary Report: Cyber Attack on WazirX Multisig Wallet.

Available at: https://wazirx.com/blog/preliminary-report-cyber-attack-on-wazirx-multisig-wallet/

2. WazirX. (21 July 2024). Important Update: Cyber Attack Incident and Measures to Protect Your Assets.

Available at: https://wazirx.com/blog/important-update-cyber-attack-incident-and-measures-to-protect-your-assets/

3. WazirX. (25 July 2024). WazirX Cyber Attack: Key Insights and Learnings.

Available at: https://wazirx.com/blog/wazirx-cyber-attack-key-insights-and-

learnings/

4. WazirX. (28 July 2024). Day-Wise Report: WazirX Cyber Attack Incident.

Available at: https://wazirx.com/blog/wazirx-cyber-attack-day-wise-report/

5. WazirX. (6 August 2024). Important Update: FIR Registered Regarding Cyberattack.

Available at: https://wazirx.com/blog/update-fir-registered/

6. WazirX. (18 July 2025). One Year Since WazirX Cyberattack: Rising, Rebuilding, and Returning Stronger.

Available at: https://wazirx.com/blog/one-year-since-wazirx-cyberattack-rising-rebuilding-and-returning-stronger/

7. Additional news sources: Include the specific newspaper or news articles you actually read for the case study. Record each article's author, publication date, headline, and URL where available.

DISCLAIMER

This case study titled "WazirX Cyber Attack: A Cybersecurity Case Study (2024)" has been prepared for academic purposes as part of my BCA course.

The information presented in this report has been collected from publicly available news articles, official statements, reports, and other online sources. Every effort has been made to present the information clearly and accurately based on the available sources.

This report is intended only for educational purposes and does not represent an official investigation or independent forensic analysis of the cyber attack. Certain technical details and findings are based on the information publicly reported by WazirX and other sources at the time of writing.

The views and explanations presented in this report are intended to help understand the incident from a cybersecurity perspective. The report does not intend to make

unsupported allegations against any individual or organization. The original sources are acknowledged in the References section.

This case study has been prepared to understand the importance of cryptocurrency security, cyber threat prevention, incident response, and the protection of digital assets.

THANK YOU